INCH-POUND

MIL-P-87993 (84)

2 Feb 90

MILITARY SPECIFICATION

PAVING MACHINE
RUBBER TIRED, BITUMINOUS MATERIAL, SELF PROPELLED

This specification is approved for use within WR-ALC Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers a rubber tired, bituminous material, self-propelled paving machine.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications ,standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
VV-F-800	Fuel Oil, Diesel

Military

MIL-P-514	Plates, Identification, Instructions and Marking, Blank
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-23377	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-P-26915	Primer Coating, Zinc Dust Pigmented, for Steel Surfaces

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRG, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3895

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MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies) Aluminum and Aluminum Alloys
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-S-81733	Sealing and Coating Compound, Corrosion Inhibitive
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic, Hydrocarbon Base, Aircraft
MIL-C-83286	Coating, Urethane Aliphatic Isocyanate, for Aerospace Applications

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-12	Abbreviations for Use on Drawings and on Specifications, Standards, and Technical Documents
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-721	Definition of Terms for Reliability and Maintainability
MIL-STD-808	Finishes, Materials, and Processes for Corrosion Prevention and Control in Support Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-965A	Parts Control Program
MIL-STD-970	Standards and Specifications, Order of Preference for the Selection of
MIL-STD-1223	Nontactical Wheeled Vehicle Treatment, Painting, Identification Marking, and Data Plate Standards
MIL-STD-1556B	Government/Industry Data Exchange Program (GIDEP) Contractor Participation Requirements
MIL-STD-1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapon Systems
MIL-STD-1587	Materials and Process Requirements for Air Force Weapon Systems
MIL-STD-1595	Qualification of Aircraft, Missile and Aerospace Fusion Welders
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

Handbooks

MIL-HDBK-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues and those cited in the solicitation.

US Air Force (USAF) Drawings

8838302	Fastener Quality Assurance Requirements for WR-ALC/MMV Vehicle and Support Equipment

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are Department of Defense (DoD) adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be those in effect on the date specified in the solicitation.

American Society of Mechanical Engineers (ASME)

Welding Qualifications – Section 9-86

(Application for copies should be addressed to American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York NY 10017.)

American Welding Society (AWS)

Welding Qualifications – B2.1-84

(Application for copies should be addressed to American Welding Society, 2501 NW Seventh Street, Miami, FL 33125.)

Society of Automotive Engineers (SAE)

J293	Vehicle Grade Parking Performance Requirements
J534	Lubrication Fittings
J992b	Brake System Performance Requirements - Truck, bus, and Combination of Vehicles
J1349	Engine Power Test Code - Spark Ignition and Diesel

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification; however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Design.

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities, and services not specified herein shall be selected in accordance with MIL-STD-970.

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3.1.2 Description. The paver shall be self-propelled rubber tired mounted and shall lay, compact, and finish bituminous concrete strips. The paver shall consist essentially of a receiving hopper, spreader, compacting unit, and screed with extension and cutoff shoes.

3.1.3 Materials. The selection of materials and processes shall be in accordance with MIL-STD-1568 and MIL-STD-1587. Materials not covered by this specification or referenced document shall be suitable for the intended environment and application. The contractor shall not use nonstandard or proprietary materials and processes without prior government approval. The components, pieces and parts incorporated in the paving machine may be newly fabricated from recovered materials to the maximum extent practical, provided the paving machines produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the paving machine. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinquished from virgin raw materials.

3.1.3.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used. Surface preparation, finishes and coatings shall be in accordance with MIL-STD-808. The contractor shall prepare and submit for government review, a corrosion prevention and control plan (CPCP). The CPCP shall include, as a minimum, the contractor's methodology for incorporating corrosion prevention and control at the design level and a surface preparation and finish specification for all materials in accordance with MIL-STD-1568, paragraphs 5.1.1 and 5.1.2. Submittal of above documents does not constitute release from any specification requirements.

3.1.3.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.3.3 Primers. MIL-P-23377 and MIL-P-26915 epoxy primers shall be used on aluminum and ferrous alloys respectively.

3.1.3.4 Topcoat. MIL-C-83286 polyurethane shall be used as the topcoat finish.

3.1.3.5 Drainage. Complete drainage shall be provided for both internal and exterior pockets, areas, parts and components.

3.1.3.6 Magnesium. Magnesium and magnesium alloys shall be subject to approval by the government.

3.1.3.7 Prohibited materials. The use of wood and/or wood products is prohibited. Similarly, vinyl, polyvinyl chloride, polyester, and corrosive type RTV (yields acetic acid upon curing) are prohibited for use in all forms, interior and exterior.

3.1.3.8 Fasteners. Ferrous alloy fasteners shall be cadmium plated per QQ-P-416 Type II, to a minimum thickness of 0.0003 inches unless threaded tolerances dictate a different thickness; and aluminum alloy fasteners shall be anodized per MIL-A-8625. Fastener installation shall be in accordance with MIL-STD-1568.

3.1.3.9 Sealants for permanent fasteners. All joints, rivets and mating/faying surfaces of exterior and interior components shall be sealed using MIL-S-81733.

3.1.3.10 Conformal coating. Printed circuit/printed wire boards shall be conformal coated with materials in accordance with MIL-I-46058.

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3.1.3.11 Commercial off-the-shelf equipment. All commercial equipment and parts shall be assessed by the contractor as to their suitability to function in the intended environment and comply with the system's life expectancy requirements.

3.1.3.12 Galvanized steel. Many producers of zinc-coated steel apply inhibitors to protect their products from moisture during shipment and storage. Some of these inhibitors interfere with pretreatments resulting in poor priming surfaces. Contractors shall consult with their supplier of galvanized steel for recommendations on the pretreatment to be used on their respective product.

3.1.4 Design and construction. The paver shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations, and other conditions incident to operation, shipping, and storage with minimum loss of time for maintenance, repair, and servicing. The design of the vehicle shall ensure safe, efficient, and economical operation. All assemblies, controls, and installed equipment shall be located so that there is no adverse interference with each other, or with the operation, and shall be readily accessible for maintenance, operation, and replacement.

a. Finishes. Protective coating application shall be considered during the design stage. Sharp, angular corners are undesirable due to coating failures on outside corners (too thin) and the tendency of coatings to bridge inside corners. Smooth, round corners allow uniform coating thickness and avoids creation of possible sump areas. All mechanically joined components shall be finished before assembly. Welded structures shall have finishes applied after welding. When welded and bolted assemblies are used, manufacturing steps shall be arranged to minimize disruption of the protective coatings. Supplementary corrosion prevention treatments - rustproofing, sealants, greases, etc., shall be applied after finish application. All metallic surfaces shall be treated for corrosion prevention.

b. Joints and faying surfaces. Joint designs shall minimize crevices formed by overlapping surfaces and shall point away from the forward direction of vehicle travel or point down. Butt welds are preferable to lap welds. Continuous welds are preferable to tack welds. The joints shall be designed so there are no gaps on joint laps. This can be accomplished by continuous welds, solder or sealant bead lines. All faying surfaces shall be primed and completely coated with sealant prior to assembly.

c. Core panel structures. Pretreat and prime all internal surfaces prior to assembly. Pretreat and prime interior panel skin and faying surfaces prior to assembly. Coat entire faying surface area with sealant during assembly and apply sealant bead after assembly to the lap joint. The panels shall then be foamed in place using non-reverting polyurethane.

d. Alternate materials. Use of composites, polymers, etc., type materials should be investigated. While not subject to corrosion, consideration of material interaction, environmental stability and field repair should be investigated before recommending alternate material.

3.1.4.1 Reliability. The paver, including all components, shall have an observed Mean-Time-Between-Failure (MTBF) of not less than 100 hours. The paver shall also have an operational availability of not less than 0.99. A failure is defined as the event or inoperable state in which any item or part of an item does not, or would not, perform as previously specified. Operational availability (Ao) is defined as the percent of time that the system is capable of satisfactorily performing in the operational environment. Mathematically, Ao is expressed as:

Ao = MTBF/(MTBF + MRT); where MRT is Mean Repair Time

Mean repair time (MRT) is the elapsed time to accomplish all repairs divided by the number of repairs performed. Mathematically, MRT is expressed as:

MRT = T/N; where T is Elapsed Time
N is Number of Repairs.

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The elapsed time includes all maintenance actions required to correct the malfunction, including time to prepare, remove, replace, repair, adjust, and check. MRT does not include administrative or supply delays. Note that time to troubleshoot is not included; thus, this is not the standard definition for MRT.

Definitions of reliability terms not included herein shall be as found in MIL-STD-721 and MIL-HDBK-781. In the event of a conflict between these documents, MIL-HDBK-781 shall take precedence.

3.1.4.2 Maintainability. The paver shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.
b. A minimum amount of time for assembling, disassembling, locating trouble sources, and maintaining the paver. Where practical, components shall be located for rapid inspection and recognition of potential failure.
c. Ease of adjusting, servicing, and replacing parts and components.
d. Use of readily available standard tools and equipment for maintenance.
e. Maintenance with a minimum number of tools.
f. Maximum repair time, excluding time to troubleshoot, of no greater than eight hours.
g. Maximum time to perform all daily scheduled maintenance and servicing tasks of no greater than 15 minutes.

3.1.4.2.1 Maintenance tasks. The vehicle shall be designed so that one mechanic, using common hand tools, can perform the following maintenance operations within the time specified:

a. Remove, replace, and adjust all engine driven belts - 45 min.
b. Remove and replace alternator - 30 min.
c. Remove and replace all filters, screens, and strainers in all hydraulic systems - one hour.
d. Remove and replace engine coolant system hoses - 30 min.
e. Drain engine lubricating oil, remove, and replace oil filter element(s) and refill crankcase - one hour.
f. Remove and replace fuel filter element(s) - 30 min.
g. Disconnect battery cables, remove and replace batteries and reconnect battery cables - 30 min.
h. Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers, and refill converter and transmission - one hour.
i. Remove and replace starter - one hour.
j. Remove or install the engine/transmission package, two mechanics - four hours.

3.1.4.3 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for attachment of trim items or as provided in commercial components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter. All fasteners of Grade 5 or higher must meet the quality assurance requirements of US Air Force Drawing 8838302.

3.1.4.4 Servicing provisions. Drains, grease fittings, lines, and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.1.4.5 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.1.4.6 Hydraulic fluid. Hydraulic fluid shall be in accordance with MIL-H-83282.

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3.1.4.7 Component protection All space in which work is performed during operation, service, and maintenance of the paver shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.4.8 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the vehicle.

3.1.5 Environmental conditions. The paver shall be capable of satisfactory operation between +32°F and 100°F.

3.1.6 Weight distribution. The weight distribution of the paver in any loading condition shall not be unbalanced or improperly distributed. The difference in load between the left and right wheels of any axle shall not exceed 2.5 percent of the total load on the axle.

3.1.7 Equipment operation sound level. Unless otherwise specified (See 6.2), the sound level of the paver shall be in accordance with OSHA requirements.

3.1.8 Air transportability. The paver shall be air transportable without shoring in accordance with MIL-STD-1791 in Types C-130, C-141, and C-5 Aircraft and shall be air transportable in a Type C-17 Aircraft.

3.1.8.1 Tiedowns. The paver shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport. The attachment points shall be designed for use with attachment in the aircraft as described in MIL-STD-1791. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure. The tiedown points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook.

3.9 Operators platform and controls. An operator's platform shall be provided and shall be located so that the operator will have the optimum view of all operations. The platform shall be of steel with either a raised pattern finish anti-skid surface or a non-skid covering of a bonded adhesive material. All controls (except the screed controls) necessary for operation shall be provided, shall be accessible to the operator, and shall be located so that the operator can control all operations (except speed range control) from both the right and left sides of the platform. All controls shall be identified. An operator's seat with a backrest shall be provided. The seat and the backrest shall be padded and shall have waterproof coverings. Provisions shall be made to mount the seat on the right or on the left side of the operator's platform.

3.9.1 Hopper. The receiving hopped shall have a capacity of not less than 4.5 cubic yards and shall receive material from dump trucks. The hopper shall have a dumping clearance of not more than 24 inches above the ground surface, and shall be equipped with adjustable gates to control the flow of material to the spreading mechanism. The side walls of the hopper shall lower and raise hydraulically to provide clearance for the low dumping trucks and provide a positive material flow to completely empty the hopper of all material. The front end of the hopper shall have an adjustable apron or other means to prevent material from spilling off the hopper. The front end of the hopper shall be equipped with a push-bar to push a dump truck. The push-bar shall be pivotally mounted and shall be equipped with steel rollers and hydraulic truck hitch. The hopper shall have dual-bar feeders with variable control to deliver material to the spreader screws. Each side of each feeder shall operate independently of the other side.

3.9.2 Compaction. The compacting unit shall vibrate at not less than 2,000 cycles per minute. Compacting-unit extensions equal in length to screed extensions shall receive the same vibration as the compacting unit.

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3.9.3 Spreader screws. The spreader screws shall distribute the bituminous material evenly to the whole width of the compacting unit or across either half of the compacting unit as an independent operation.

3.9.4 Screed. The negative crown to not less than .5 inch, and the positive crown to not less than 1.5 inch finished material shall be controlled by a screed. The screed shall be raised and lowered by power-operated hydraulic hoist. A mechanical locking device that will hold the screed in the raised position during traveling shall be provided. A steel platform with antiskid surface shall be provided for the screed operator. The screed shall incorporate adjustments to control the thickness of the pavement strip. Sufficient adjustment shall be provided to lay pavement at any thickness between .5 inch and 8 inches.

3.9.5 Screed heater. The paver shall be provided with one or more low-pressure-air-type diesel fuel burners which shall provide even distribution of heat over the entire area of the screed and the screed extensions to prevent bitumen from adhering. The burner shall be ignited and controlled from the screed operator's platform. Fuel may be supplied from the main fuel tank or from a separate tank. A fuel pump, an air blower, a fuel sediment bowl, and fuel lines shall be furnished with the burner system.

3.9.6 Fuel sediment bowl. The sediment bowl shall be provided between the fuel tank and the burner. The sediment bowl shall be removable in such a manner as to minimize fire hazard. The contents of the sediment bowl shall be visible without the bowl being removed.

3.9.7 Fuel lines. Fuel lines shall be either steel or oil-resistant hose, fuel lines shall be supported by metal clips and secured to prevent vibration and chafing.

3.9.8 Power transmission. Primary power transmission shall consist of a hydrostatic pump capable of continuous operation at the rated horsepower of the engine and hydrostatic drive motors to drive the paver and its major components. The hydrostatic drive system shall provide for infinite variation of horsepower transfer to the drive wheels in each range of speed selection, as well as, suitable control of speed and power of other components. Chains or belts where used to drive minor components shall be the manufacturer's standard. Safety guards shall be provided for all exposed chain drives.

3.9.9 Accessories. Each paver shall be provided with screed extensions, cutoff shoes, and a spray cleaner, complete with separate fuel pump and oil resistant hose of sufficient length to clean all parts of the machine with diesel oil from the main tank.

3.9.10 Screed extensions. Screed extensions shall increase the paving width in increments of 6 inches from 8 feet to either 10 feet or 12 feet. A combination of screed extensions and cutoff shoes may be used to attain increments of 3 inches. Extensions shall be attached to the screed by bolting, pins, or by quick-coupling, when hydraulic-actuated vibration is employed. Screed extensions shall be provided for the compacting unit as well as for the oscillating screed.

3.9.11 Cutoff shoes. Cutoff shoes shall reduce the paving width in increments of 3 inches from 8 feet to 6 feet.

3.10 Performance.

3.10.1 Mobility. The fully loaded and equipped paver shall:

a. Maintain a paving speed of not less than 11 feet per minute (FPM) minimum and not less than 150 FPM maximum. The maximum forward and reverse travel speed shall be not less than 10 mph.
b. Ascend and descend a 20.4 percent grade without spillage of any fluid.
c. Negotiate a 20 percent side slope in each direction.
d. Negotiate, in each direction, a ramp with a 17° minimum slope connecting two horizontal surfaces. The ramp length shall be at least 1.2 times the wheelbase of the paver. The paver shall be driven forward or backward from one horizontal surface across the ramp crest, ascend or descend the

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ramp, cross the ramp crest and proceed on the other horizontal surface without any portion of the paver, other than its tires, contacting any of the three surfaces.
e. Comply with all service brake performance requirements of SAE J992b.
f. Comply with all parking brake performance requirements of SAE J293, except that the required grade shall be 20.4 percent.

3.11 Powertrain

3.11.1 Engine. The paver shall be equipped with a commercial liquid cooled diesel engine of not less than six cylinders. The total power output required by all items driven by the engine shall not exceed its net power output rating at 2,000 feet elevation, determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel in accordance with VV-F-800 and shall operate satisfactorily on Grades JP-4 and JP-5 turbine fuel in accordance with MIL-T-5624 and Grade JP-8 in accordance with MIL-T-83133. Any engine that is adversely affected by any of the above fuels is not acceptable.

3.11.1.1 Engine cooling system. The engine shall be equipped with a cooling system capable of satisfactory performance under the environmental conditions specified herein; coolant temperature shall not exceed the manufacturer's recommended maximum. Expansion tanks of sufficient capacity for expansion of all coolant in the system shall be provided.

3.11.1.2 Exhaust system. The exhaust system shall be designed to expel exhaust gases in a manner that is not hazardous to operators or mechanics; the outlet shall not be directed toward the ground. Th exhaust system, including all fasteners, shall be constructed of corrosion resistant materials. The system shall be protected from damage which could result from traversing rough terrain.

3.11.1.3 Transmission. A commercial automatic transmission shall be provided, complete with a hydraulic torque converter. The normal drive range selector position shall provide not less than four gear ratios without selector movement. The transmission shall have a torque capacity of at least 105 percent of the maximum net engine torque.

3.11.2 Steering system. The paver shall be equipped with a power assisted steering system, with manual steering available in the event of power assist failure. The steering system shall not require more than four turns lock to lock.

3.11.3 Brakes.

3.11.3.1 Service brakes. The paver shall be equipped with full air brakes on all wheels. Brakes shall be of either drum or disc type. A commercial air dryer and necessary associated parts shall be added to the pneumatic system.

3.11.3.2 Parking brakes. The paver shall be equipped with parking brakes of the spring applied type.

3.11.4 Wheel suspension. Large high flotation, heavy duty sand service pneumatic tires, hydraflated to provide maximum tractive effort shall be provided on the rear of the machine. Small diameter solid rubber tired tandems bogie wheels for maximum flotation and equal weight distribution shall be provided on the front of the machine.

3.11.5 Fuel system.

3.11.5.1 Fuel tank. A fuel tank shall be provided with a capacity of not less than eight hours capacity operation. The tank shall have a fill opening of three inches minimum readily accessible to personnel standing on the ground. It shall be located and mounted so as to provide maximum protection from damage.

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3.11.5.2 Fuel filter. At least one filter of the fuel water separator type shall be located in the fuel line between the fuel tank and the engine. It shall be installed so that the element can be readily serviced without removing surrounding equipment. Unless the filter is mounted higher than the fuel tank, a shutoff valve or check valve shall be included in the inlet line to the filter.

3.12 Electrical system. The paver shall be equipped with a 12 volt electrical system; a 24 volt starting system may be provided. Wiring shall be weatherproof and color coded or numbered for easy identification. All circuits shall be protected by circuit breakers located so as to be protected from the environmental conditions described herein and to be readily accessible to a mechanic.

3.12.1 Alternator. The engine shall be equipped with an alternator sufficient for operating all paver electrical components simultaneously. Minimum capacity of the alternator shall be 60 amperes.

3.12.2 Batteries. The paver shall be equipped with at least two 12 volt batteries, rated at a total of 1200 cold cranking amps (CCA) minimum. Batteries shall be so arranged and connected to provide sufficient capacity for engine starting in all conditions specified herein. The batteries shall be mounted in an acid resistant tray; a sliding carriage shall be provided if the batteries must be moved sideways to permit access for service or replacement. A master switch shall be provided to disconnect the batteries from the electrical system.

3.13 Winterization system. The paver shall be equipped with a winterization system to permit storage in ambient temperatures as low as -40°F without detrimental effect on the paver.

3.14 Parts control. The contractor shall participate in Government Industry Data Exchange Program (GIDEP) in accordance with MIL-STD-1556B. In the design of the paver, the contractor shall ensure the components selected have not been identified as a problem or unresolved deficiency by GIDEP. The contractor shall submit an Alert/Safe-Alert, on a DD Form 1939, whenever an actual or potential problem is evident with respect to a part, component, material, test equipment, process or safety condition. The contractor shall certify maximum use of off-the-shelf items in accordance with MIL-STD-965A.

3.15 Finish and protective coatings.

3.15.1 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808, except that commercial components may be prepared in accordance with the manufacturer's standard practice, provided it is compatible with the exterior finish paint.

3.15.2 Finish. The exterior paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion, or other defect and shall be capable of passing the test described in paragraph 4.8.10 of MIL-C-83286.

3.15.2.1 Forest Green. Unless otherwise specified (see 6.2), the outer surface of the paver and the interior surfaces of compartments shall be painted with polyurethane paint, Forest Green, Color Number 24052 of FED-STD-595. The paint shall be in accordance with MIL-C-83286, or shall be a commercial polyurethane paint which meets the following requirements of MIL-C-83286:

(1) Paragraph 3.7.3.4 Low temperature resistance.
(2) Paragraph 3.7.3.6.1 Accelerated weathering.
(3) Paragraph 3.7.3.7 Resistance to taping.

In addition, the paint shall have a minimum pencil hardness of 6H when tested in accordance with paragraph 4.8.12 of MIL-C-83286. All exterior trim items (such as wheels, wheel covers, and light trim rings) shall be painted with the same paint as the exterior of the paver.

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3.15.2.2 Desert Sand. When specified (see 6.2), the outer surface of the paver and the interior surfaces of compartments shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30313 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, and light trim rings) shall be painted with the same paint as the exterior of the paver.

3.15.3 Rustproofing. Unless otherwise specified (see 6.2), the paver shall be rustproofed in accordance with the contractor's standard commercial practice. The first article units shall not be rustproofed until after approval of the first article test report.

3.16 Markings. Unless otherwise specified, markings for vehicles painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595, and markings for vehicles painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.16.1 Identification markings. Unless otherwise specified (see 6.2), the paver shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.17 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc., shall conform to MIL-P-514 and shall be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.17.1 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the paver in a readily accessible location. The identification shall contain the following information:

NOMENCLATURE
MANUFACTURER'S MODEL NUMBER
MANUFACTURER'S SERIAL NUMBER
MANUFACTURER'S NAME
MANUFACTURER'S ADDRESS
MANUFACTURER'S SERVICE PHONE NUMBER
DATE OF MANUFACTURE
CONTRACT NUMBER
NATIONAL STOCK NUMBER

3.17.2 Control markings. All controls, valves, gauges, and indicators used in the operation of the paver shall be identified by nameplates securely installed in clearly visible locations.

3.17.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise, and adequate to enable operation of the paver without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.17.4 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the paver.
b. Horizontal and vertical location of the center of gravity of the paver in transport condition.
c. Shipping weight.
d. Loading cubage.
e. Overall length, width, and height.
f. Tiedown instructions for air transport in Types C-130, C-141, C-17, and C-5 Aircraft.

3.18 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be marked for identification in

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accordance with MIL-STD-130. (Note that MIL-STD-130 does not require marking of commonly known commercial parts which present no identification problem or parts in assemblies which are not normally subject to disassembly or repair.)

3.19 Dimensions. The paver, in fully operational condition, including all equipment and accessories, shall not exceed the following dimensions:

a. Length, 240 inches.
b. Width, 108 inches.
c. Height, 101 inches.

3.20 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2.1).

3.2.1 Fabrication.

3.21.1 Welders. All welders employed in the fabrication of the paver, except for commercial components, shall be certified in accordance with the applicable requirements of MIL-STD-1595, the American Welding Society, or the American Society of Mechanical Engineers.

3.21.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings. Ferrous and aluminum welding shall be in accordance with MIL-STD-1595.

3.21.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.21.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.21.5 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of backlash.

3.21.6 Cleaning. The paver shall be cleaned. Loose, spattered, or excess solder, welding slag, stray bolts, nuts and washers, rust, metal particles, pipe compound, and other foreign matter shall be removed during and after final assembly.

3.21.7 Soldering. Soldering shall be in accordance with MIL-STD-454, Requirement 5.

3.21.8 Brazing. Brazing shall be performed in accordance with MIL-B-7883.

3.22 Workmanship. The paver, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; marking of parts and assemblies; alignment of parts and tightness of assembly fasteners; etc. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor' s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the paver shall be classified as follows:

a. First article testing (see 4.4 and 6.2.1).
b. Quality conformance inspection (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.
b. Time started.
c. Time finished.
d. Ambient temperature.
e. Engine speed.
f. Engine oil pressure.
g. Engine coolant temperature.
h. Transmission oil temperature.
i. Engine hourmeter reading at the start and finish of each test.
j. Hydraulic fluid pressure at the pump outlet.
k. Hydraulic fluid temperature in the reservoir.

4.3.3 Preparation for tests. The paver, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all test specified herein, the paver shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.
b. Spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any operating condition, except as allowed herein.

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c. Structural failure of any component, including permanent deformation, or evidence of impending failure.
d. Evidence of excessive wear.
e. Interference between vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.
f. Misalignment of components.
g. Conditions which present a safety hazard to personnel during operation, service, or maintenance.
h. Overheating of the engine, transmission, or any other vehicle component.

4.4 First article testing.

4.4.1 First article test sample. The first article test sample shall be identical to the items procured under the terms of the contract. Approval of the first article test report shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.2 First article tests. The first article tests shall consist of all tests listed under 4.6.

4.4.3 Test report. Upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as a part of the test report:

a. All data collected, failures, maintenance performed, and other events recorded shall be identified by accumulated operation time, miles, cycles, or position in the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.
b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.
c. Summary of the engineering reasoning behind any corrections made, to be made, or proposed to be made and the predicted effectiveness of those corrections.
d. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808.
e. List of special tools and diagnostic equipment necessary to perform field level maintenance.

4.5 Quality conformance inspection.

4.5.1 Examination. Each paver shall be examined for the following characteristics. Presence of one or more defects shall be cause for rejection.

101. Dimensions not as specified.
102. Material not as specified.
103. Parts or components not as specified.
104. Controls not accessible from operator's platform.
105. Lubrication not as specified.
106. Lifting or tiedown attachments not as specified.
107. Identification markings or other markings not as specified.
108. Treatment or painting not as specified.
109. Engine or engine components not as specified.
110. Workmanship not as specified.
111. Hopper not as specified.
112. Compaction not as specified.
113. Spreader screws not as specified.
114. Screed not as specified.
115. Screed heater not as specified.
116. Fuel sediment bowl not as specified.

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117. Fuel lines not as specified.
118. Transmission not as specified.
119. Screed extensions and cutoff shoes not as specified.

4.6 Test methods.

4.6.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting, and markings. Any painted surface on the exterior of the vehicle may be tested in accordance with paragraph 4..8.10 of MIL-C-83286.

4.6.2 Mobility tests.

4.6.2.1 Grade operation. The fully loaded paver shall be driven up and down a 20.4 percent dry paved grade. The maximum speed that can be maintained while climbing the grade shall be measured and recorded in the test report. Engine oil pressure shall be observed and shall not be less than the manufacturer's recommended minimum pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the grade.

4.6.2.2 Ramp test. The fully loaded paver shall be driven forward from a hard flat horizontal surface and ascend a ramp with a minimum length of 1.2 times the vehicle wheelbase and a slope of 17°. It shall then proceed across the ramp crest onto an upper connecting hard horizontal surface until all tires are resting on the upper surface, down the ramp, and onto the lower horizontal surface. The test shall then be repeated in the opposite direction.

4.6.2.3 Dimensional verification. The dimensions specified herein shall be measured with the vehicle standing on a flat level surface. Overall height and paver lean shall be measured in both the empty and the fully loaded conditions.

4.6.3 Sound level testing. The paver shall be tested in accordance with OSHA requirements.

4.6.4 Operational checkout. All controls, levers, and mechanisms shall be actuated a sufficient number of times to ensure consistency of operation and determine whether the test exposure produced system degradation.

4.6.5 Maintainability demonstration. Compliance with the repair time requirements shall be demonstrated by task performance. All scheduled maintenance tasks and all tasks listed in 3.1.4.2.1 shall be performed. In addition, corrective maintenance tasks shall be selected by the procuring activity from an approved list of tasks provided by the contractor in the first article test procedure. This list shall represent all expected failures over the life of the paver. Not less than 25 or more than 50 of the tasks shall be demonstrated to verify a maximum repair time within the required limit. As part of this demonstration, the recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance time per measure of use, such as calendar time, hours of operation, or miles traveled.

4.6.6.1 Static. Raise and block the paver so that the wheels, screed (and extensions), and other operating components are off the ground or place the paver on a lubricated steel plate. Start the engine and set all operating components in motion and operate at low speed for not less than 1 hour. Manipulate all controls, ignite the screed burner and operate for 5 minutes. Inability to manipulate all controls, excessive heating of gear cases, bearings, and bushings, or uneven distribution of heat over the entire screed area shall constitute failure of this test.

4.6.6.2 Crown. Adjust the screed to the maximum negative crown. The adjust to the maximum positive crown. Nonconformance to 3.9.4 shall constitute failure of this test.

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4.6.6.3 Paving width. Using the cutoff shoes, reduce the paving width in the specified increments through the range specified in 3.9.10. Then using the screed extensions or a combination of screed extensions and cutoff shoes, extend the screed in the specified increments through the range specified in 3.9.9. Nonconformance to 3.9.9 or 3.9.10 shall constitute failure of this test.

4.6.6.4 Flywheel horsepower test. Compliance with the manufacturer's data listing the applied net horsepower at the flywheel shall be determined from the manufacturer's certified power curve, verified for diesel engines by regular production dynamometer tests, with standard allowances for friction variations in new engines and power deductions for fan, alternator and water pump if not on engine during test. Failure to comply shall be cause for rejection.

4.6.6.5 Paving and travel speeds. Operate the paver on an improved level surface and determine the following:

(a) Minimum paving speed.
(b) Maximum paving speed.
(c) Maximum forward travel speed.
(d) Maximum reverse travel speed.

Nonconformance to 3.10.1 shall constitute failure of this test.

4.6.7 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.1.9. The analysis shall include the tiedowns and all major components and their ability to withstand the accelerations specified in MIL-STD-1791. The evaluation shall also include a dimensional analysis for the vehicle while loaded aboard Types C-130, C-141, C-17, and C-5 Aircraft.

4.7 Refurbishment. After successful completion of the all required testing, each vehicle shall be restored to like new condition.

5. PREPARATION FOR DELIVERY.

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with the terms of the contract.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The paver is intended to receive mixed bituminous material from trucks and to lay the bituminous material for road and airfield construction.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
   Finish color required, if not Forest Green, Color Number 24052 of FED-STD-595 (see 3.15.2).
   Rustproofing required, if other than the manufacturer's standard commercial practice (see
   Identification markings required (see 3.16.1).
b. Sound level testing other than specified (see 4.6.3).

6.2.1 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of the first article. Invitations for bids should provide that the Government reserves the right to

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waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Data Requirements. The following Data Item Descritpions (DIDs) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

DID Number	DID Title	Suggested Tailoring
DI-T-4901	FAT Procedures	See DD Form 1423
DI-T-4902	FA Inspection Report	See DD Form 1423
DI-T-2373A/T	Notification of Test	See DD Form 1423

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Forms 1423.

6.4 Subject term (key word) listing.

Compacting unit
Cutoff shoes
Hopper
Screed
Sediment bowl
Spreader screw

Custodians:
Air Force - 99

Preparing activity:
Air Force - 84

Project No: 3895-F251

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-87993	Paving Machine, Rubber Tired, Bituminous Material, Self Propelle

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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